



Attacks on AI Infrastructure

AI systems have become valuable targets for attackers who exploit vulnerabilities in data, models, and infrastructure. Defending your AI systems begins with understanding these sophisticated threats.

Why Target AI?

Intellectual Property

AI models represent significant R&D investment and valuable IP, making them prime targets for theft

Sensitive Data

Training datasets often contain PII, medical records, financial data, and other confidential information

Resource-Rich Infrastructure

GPU clusters and computing resources can be hijacked for cryptocurrency mining operations

Compromise Impact:

- Financial losses
- Reputation damage
- Regulatory penalties
- Competitive disadvantage
- Customer harm

Attack Surfaces in AI Systems

1

Data Pipelines

Poisoning attacks, manipulation of training sets, and injection of malicious samples during data collection

2

Models

Extraction of weights, model inversion to recover training data, and adversarial examples to manipulate outputs

3

Infrastructure

GPU theft, API abuse, insider threats, and exploitation of MLOps vulnerabilities

4

Deployment Endpoints

API scraping, query flooding, and inference manipulation through carefully crafted inputs

Each attack surface requires specific security controls

Data-Centric Attacks



→ Data Poisoning

Malicious samples injected during training to compromise model behavior

→ Backdoors

Hidden triggers embedded in training data that cause predictable malicious outputs

→ Label Flipping

Corrupting ground truth labels to reduce model accuracy in targeted ways

⊗ **Risk: Compromised model from the ground up** - These attacks are particularly dangerous because they compromise the foundation of the AI system

Model-Centric Attacks

Model Extraction

Attackers systematically query an API to reconstruct model architecture and weights, effectively stealing the model

Example: Querying a classification API thousands of times to map decision boundaries

Model Inversion

Attackers reverse-engineer models to recover sensitive training data that was used to build the model

Example: Reconstructing faces from a facial recognition model

Membership Inference

Determining whether specific data points were used during model training, revealing sensitive information

Example: Discovering if a patient's medical record was in a training dataset

Adversarial Examples

Specially crafted inputs with imperceptible perturbations that cause models to make incorrect predictions

Example: Adding noise to a stop sign image that makes it classified as a speed limit sign

Infrastructure-Centric Attacks

GPU/TPU Hijacking

Compromised access credentials or container escape attacks lead to compute theft for crypto mining operations

Kubernetes Misconfigurations

Improperly secured K8s clusters expose data pipelines and model artifacts to exfiltration

Insider Attacks

Privileged users compromise MLOps pipelines to steal models or inject backdoors

Dependency Attacks

Poisoned PyPI/conda packages introduce malicious code into training environments



High-value GPU clusters are prime targets for attackers

Deployment Attacks



API Abuse

Excessive queries to extract model functionality or exhaust resources, potentially leading to service disruption



Denial-of-Service

Deliberately overloading inference endpoints to cause service outages or increased latency



Adversarial Queries

Carefully crafted inputs designed to trick production models into making specific incorrect predictions



Shadow AI Models

Stolen models deployed by attackers, potentially offering similar services or intercepting legitimate traffic



Why These Attacks Matter

87%

High-Risk Industries

Percentage of healthcare and autonomous vehicle companies where AI safety failures could lead to physical harm or death

\$1.2M

Average Recovery Cost

Cost to remediate an AI security incident, including incident response, retraining, and legal expenses

\$20M

GDPR Penalties

Maximum potential fine for data protection violations if AI systems leak PII from training data

312%

Cost Increase

Average spike in cloud infrastructure costs when GPU resources are compromised for cryptocurrency mining

 Beyond immediate financial impact, AI security failures can permanently damage customer trust and brand reputation

Threat Modeling for AI Infrastructure



Identify Assets

- Training datasets
- Model weights & architecture
- Computing infrastructure
- API endpoints
- Feature stores



Identify Adversaries

- External hackers
- Malicious insiders
- Competitors
- Nation-state actors
- Automated bots



Analyze Attack Vectors

- Access points
- Authentication weaknesses
- Data pipeline vulnerabilities
- Infrastructure misconfigurations



Build Defenses

- Differential privacy
- Model encryption
- Continuous monitoring
- RBAC & zero trust

Systematic threat modeling reveals vulnerabilities before attackers can exploit them

AI Security Framework

Data

Poisoning, label-flipping attacks

Infrastructure

GPU hijacking, dependency attacks



Model

Extraction and inversion threats

Deployment

API abuse and adversarial queries

AI infrastructure is only as strong as its weakest link

Summary: Defending Your AI Infrastructure

Comprehensive Protection Required

AI infrastructure is vulnerable across the entire stack: data, models, infrastructure, and deployment stages

Diverse Attack Methods

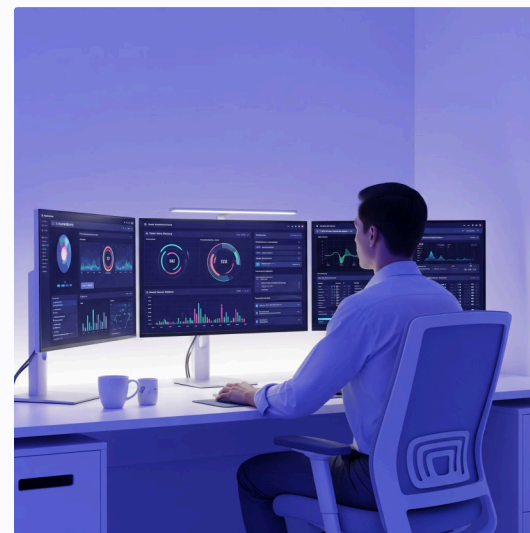
Attacks range from subtle data poisoning to model extraction to outright infrastructure abuse and API exploitation

Proven Risk

Real-world incidents demonstrate that these threats are actively being exploited in production environments

Defense Foundation

Understanding attack surfaces is the critical first step in building effective defense strategies



Next steps: Implement continuous monitoring across your AI security stack